

# Code changes from implicit to CEL

fe-large-displacement

Each script in `examples/` is complete. Open two and diff. This note lists the extra lines at each rung.

## 1 Implicit

`examples/implicit.py`

### Geometrically nonlinear static step

Automatic incrementation. No stabilisation (that is fictitious damping).

```

124 # *Static, geometrically nonlinear. Automatic incrementation. No
125 # stabilisation: that is fictitious viscous damping.
126 model.StaticStep(
127     name='Push', previous='Initial', nlgeom=ON, timePeriod=1.0,
128     initialInc=0.01, minInc=1.0e-8, maxInc=0.5)

```

### Displacement control

```

160 model.DisplacementBC(
161     name='Push', createStepName='Push', region=asm.sets['IndRP'],
162     u1=0.0, u2=0.0, u3=-IND_DEPTH, ur1=0.0, ur2=0.0, ur3=0.0,
163     amplitude='Ramp', distributionType=UNIFORM)

```

### Standard element library

```

101 elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=STANDARD)
102 for part in (soil, indenter):

```

### Standard general contact

```

171 contact = model.ContactStd(name='GeneralContact', createStepName='Initial')
172 contact.includedPairs.setValuesInStep(stepName='Initial', useAllstar=True)
173 contact.contactPropertyAssignments.appendInStep(
174     stepName='Initial', assignments=((GLOBAL, SELF, 'Interface'),))

```

## 2 Implicit ALE

On top of implicit. `examples/implicit_ale.py`

### Restricted adaptive domain

ALE only accepts first-order reduced-integration solids (C3D8R). An empty set is silent. Standard has no initial mesh sweeps.

```

169 # --- ALE (the only extra block versus implicit.py) -----
170 # Restrict the domain to a box around the indenter. An empty set is silent:
171 # Abaqus will not warn, and ALE will do nothing.
172
173 half = 3.0 * IND_HALF
174 zmin = max(-1.3 * IND_DEPTH, -SOIL_DEP + 2.0 * SEED)
175 ale_elems = soil_inst.elements.getByBoundingBox(
176     xMin=-half, xMax=half, yMin=-SOIL_WID, yMax=SOIL_WID,
177     zMin=zmin, zMax=0.0)
178 if len(ale_elems) == 0:
179     raise RuntimeError('ALE box is empty. ALE would run and do nothing.')
180 asm.Set(name='ALE_Box', elements=ale_elems)
181 print('ALE elements: %d' % len(asm.sets['ALE_Box'].elements))
182
183 model.AdaptiveMeshControl(name='ALE_BC', smoothingPriority=UNIFORM)
184 # Standard has no initial mesh sweeps. Do not pass initialMeshSweeps.
185 model.steps['Push'].AdaptiveMeshDomain(
186     region=asm.sets['ALE_Box'], controls='ALE_BC',
187     frequency=10, meshSweeps=1)

```

### Enhanced hourglass for Standard ALE

Standard ALE on C3D8R requires hourglass=ENHANCED. CAE cannot write \*Section Controls. \*Diagnostics, adaptive mesh= is Explicit-only and is a fatal unknown parameter on \*Static.

```

192 def _inject_hourglass(inp_path):
193     """Standard ALE requires enhanced hourglass on C3D8R. CAE cannot write
194     *Section Controls. *Diagnostics, adaptive mesh= is Explicit-only."""
195     lines = open(inp_path).readlines()
196     out, inserted = [], False
197     for line in lines:
198         out.append(line)
199         if (not inserted) and line.upper().startswith('*PREPRINT'):
200             out.append('*Section Controls, name=SoilSC, hourglass=ENHANCED\n')
201             inserted = True

```

## 3 Explicit

On top of implicit, replacing the solver. `examples/explicit.py`

### Mass (\*Density)

Required. The stable increment is  $L_{\min}/c_d$ .

```

80 # *Density is required. The stable time increment is L_min / wave_speed.
81 soil_mat = model.Material(name='SoilMat')
82 soil_mat.Density(table=((SOIL_RHO,)),)

```

### Explicit element library

```

98 # Same C3D8R, Explicit library.
99 elem = mesh.ElementType(elemCode=C3D8R, elemLibrary=EXPLICIT)

```

### Dynamic explicit step, not static

Drop automatic stabilisation. Bulk viscosity is on by default (ALLVD). Do not add extra damping to hide kinetic energy.

```

122 # *Dynamic, Explicit. No automatic stabilisation. Bulk viscosity is on by
123 # default (watch ALLVD). Do not add extra damping to hide kinetic energy.
124 time_period = IND_DEPTH / float(IND_VEL)
125 model.ExplicitDynamicsStep(

```

```

126     name='Push', previous='Initial', timePeriod=time_period,
127     nlgeom=ON, improvedDtMethod=ON)

```

## Energy history

Peak ALLKE/ALLIE below about 5 % is acceptable, below 1 % is quotable.

```

131 model.HistoryOutputRequest(
132     name='Energies', createStepName='Push',
133     variables=('ALLIE', 'ALLKE', 'ALLAE', 'ALLDC', 'ALLVD', 'ALLWK',
134               'ETOTAL'),
135     numIntervals=200)

```

## Velocity control, not displacement

```

160 # Velocity control, not displacement. Sign is negative: penetration is -z.
161 model.VelocityBC(
162     name='Push', createStepName='Push', region=asm.sets['IndRP'],
163     v1=0.0, v2=0.0, v3=-IND_VEL, vr1=0.0, vr2=0.0, vr3=0.0,
164     amplitude='Ramp', distributionType=UNIFORM)

```

## Explicit general contact

```

172 contact = model.ContactExp(name='GeneralContact', createStepName='Push')
173 contact.includedPairs.setValuesInStep(stepName='Push', useAllstar=True)
174 contact.contactPropertyAssignments.appendInStep(
175     stepName='Push', assignments=((GLOBAL, SELF, 'Interface'),))

```

## Double precision job

```

186 job = mdb.Job(
187     name=JOB_NAME, model='Model-1', type=ANALYSIS,
188     explicitPrecision=DOUBLE_PLUS_PACK, nodalOutputPrecision=FULL,
189     numCpus=1, numDomains=1, parallelizationMethodExplicit=DOMAIN,
190     multiprocessingMode=DEFAULT, memory=90, memoryUnits=PERCENTAGE)

```

## Optional mass scaling

Commented in examples/explicit.py. Semi-automatic, target  $\Delta t$ . Then re-check ALLKE.

## 4 Explicit ALE

On top of explicit. examples/explicit\_ale.py

### Adaptive domain with initial sweeps

Same box as implicit ALE. Explicit adds initialMeshSweeps=5.

```

172 # --- ALE (the only extra block versus explicit.py) -----
173
174 half = 3.0 * IND_HALF
175 zmin = max(-1.3 * IND_DEPTH, -SOIL_DEP + 2.0 * SEED)
176 ale_elems = soil_inst.elements.getByBoundingBox(
177     xMin=-half, xMax=half, yMin=-SOIL_WID, yMax=SOIL_WID,
178     zMin=zmin, zMax=0.0)
179 if len(ale_elems) == 0:
180     raise RuntimeError('ALE box is empty. ALE would run and do nothing.')
181 asm.Set(name='ALE_Box', elements=ale_elems)
182 print('ALE elements: %d' % len(asm.sets['ALE_Box'].elements))

```

```

183
184 model.AdaptiveMeshControl(name='ALE_BC', smoothingPriority=UNIFORM)
185 # Explicit has initial mesh sweeps. Default frequency=10, meshSweeps=1.
186 model.steps['Push'].AdaptiveMeshDomain(
187     region=asm.sets['ALE_Box'], controls='ALE_BC',
188     frequency=10, meshSweeps=1, initialMeshSweeps=5)

```

### Diagnostics after \*Dynamic, Explicit

Then read the .msg. Zero percent of nodes moved means the domain was defined and did nothing.

```

223 # CAE cannot write *Diagnostics. Insert it after *Dynamic, Explicit.
224 _inject_diagnostics(inp)

```

Run one CPU for a study of ALE. Nodes shared between parallel domains are nonadaptive.

## 5 Explicit CEL

On top of explicit, replacing the Lagrangian soil. `examples/explicit_cel.py`

### Eulerian part with void above the soil

Do not set the void to zero. Heave needs empty elements.

```

65
66 # Eulerian box: soil depth PLUS empty space above it for heave.
67 H = VOID
68 sketch = model.ConstrainedSketch(name='eul', sheetSize=10.0 * SOIL_LEN)
69 sketch.rectangle(point1=(-SOIL_LEN / 2.0, -SOIL_WID / 2.0),
70                 point2=(SOIL_LEN / 2.0, SOIL_WID / 2.0))
71 eul = model.Part(name='Eulerian', dimensionality=THREE_D, type=EULERIAN)
72 eul.BaseSolidExtrude(sketch=sketch, depth=SOIL_DEP + H)
73
74 # Partition at the soil surface so the filled region is a cell, not a guess.
75 if H > 0.0:
76     datum = eul.DatumPlaneByPrincipalPlane(principalPlane=XYPLANE,
77                                             offset=SOIL_DEP)
78     eul.PartitionCellByDatumPlane(datumPlane=eul.datums[datum.id],
79                                 cells=eul.cells)
79

```

### Eulerian section and EC3D8R

```

96 # Eulerian section, not a solid section. The instance name is the EVF handle.
97 eul_key = 'soilmat-1'
98 model.EulerianSection(name='EulSec', data={eul_key: 'SoilMat'})
99 eul.SectionAssignment(region=regionToolset.Region(cells=eul.cells),
100                     sectionName='EulSec')
101 model.HomogeneousSolidSection(name='IndSec', material='SteelMat')
102 indenter.SectionAssignment(region=regionToolset.Region(cells=indenter.cells),
103                           sectionName='IndSec', offsetType=MIDDLE_SURFACE,
104                           thicknessAssignment=FROM_SECTION)
105
106 eul.setElementType(
107     regions=(eul.cells,),
108     elemTypes=(mesh.ElemType(elemCode=EC3D8R, elemLibrary=EXPLICIT),))

```

### Initial volume fraction

Cells below  $z = 0$  start full. This is \*Initial Conditions, `type=VOLUME FRACTION`.

```

128 # Cells below z = 0 start full. Everything else starts empty.
129 filled = eul_inst.cells.getByBoundingBox(

```

```

130     zMin=-SOIL_DEP - 1e-6, zMax=0.0 + 1e-6)
131 if len(filled) == 0:
132     raise RuntimeError('no Eulerian cells below z = 0')
133 filled_set = asm.Set(name='SoilFilled', cells=filled)
134 model.MaterialAssignment(
135     name='Eulerian material assignment',
136     instanceList=(eul_inst,),
137     useFields=False,
138     assignmentList=((filled_set, (1.0,)),))

```

## Walls, free top

Zero normal motion on the sides and base. The top is open.

```

170 # Walls: zero normal motion on sides and base. The top is open.
171 top = H
172 for name, point, dof in (
173     ('XMIN', (-SOIL_LEN / 2.0, 0.0, (top - SOIL_DEP) / 2.0), 1),
174     ('XMAX', (SOIL_LEN / 2.0, 0.0, (top - SOIL_DEP) / 2.0), 1),
175     ('YMIN', (0.0, -SOIL_WID / 2.0, (top - SOIL_DEP) / 2.0), 2),

```

## No ALE

General contact is mandatory. There is no `*Adaptive Mesh` block.

## 6 What to check in the deck

| Rung         | The deck must show                                                    |
|--------------|-----------------------------------------------------------------------|
| Implicit     | <code>*Static, nlgeom=YES</code> , displacement BC                    |
| Implicit ALE | plus <code>*Adaptive Mesh</code> on ALE_Box                           |
| Explicit     | <code>*Density, *Dynamic, Explicit, type=VELOCITY, ALLKE</code>       |
| Explicit ALE | plus <code>*Adaptive Mesh, *Diagnostics</code>                        |
| Explicit CEL | <code>*Eulerian Section, EC3D8R, volume fraction, void, no ALE</code> |

Write the input. Read it. Then submit.